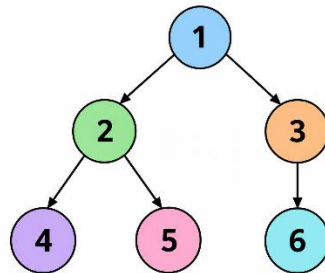


- Q1)** Write a program to implement Depth Limited Search (DLS). **[10 marks]**



- Q2)** Write a program to implement the Backward Chaining inference mechanism. **[20 marks]**

OR

- Q2)** Write a program to implement the K-Means Clustering Algorithm for grouping data into clusters. Use the data set provided in country_data.csv. The columns are: country nm, child_mortality, exports, health, imports, income, inflation, life_expectancy, total_fer, gdp. Normalize the data set for KMeans clustering (hint: use StandardScaler()). Apply KMeans with number of clusters = 3. Use different values for k from 1 to 10, use Elbow method to find the ideal value of 'k'.

- Q3)** Viva **[05 marks]**